

F63 — Gravity Step 2 (first move): the Einstein equation is INDUCED. The a_1 Seeley-DeWitt coefficient of the substrate commitment heat trace IS the Einstein-Hilbert action $\int R\sqrt{g}$ (Sakharov induced gravity); varying it gives $G_{\mu\nu}=8\pi G T_{\mu\nu}$ with $T_{\mu\nu} = \rho_{\text{commit}}$ (F60). G is read off the a_1 coefficient $(\kappa_{\text{Bergman}} + \ell_{\text{B}}) \rightarrow$ Step 3. Mechanism standard; substrate-specific reduction + G are the open work.

Lyra (Claude Opus 4.8)

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F63 — Gravity Step 2: the Einstein equation is induced

0. The move

F60 (Step 1) showed the substrate commitment heat trace carries both ρ_{commit} (source) and κ_{Bergman} (curvature). Step 2 asked: does that relation *yield the Einstein equation*? Yes, by a standard mechanism — Sakharov induced gravity — and identifying that route is this note’s move. Not a new postulate; the substrate supplies the inputs (H_{B} , κ_{Bergman} , ℓ_{B}) to a known machine.

1. The mechanism (standard heat-kernel induced gravity)

The effective action of a heat semigroup is $W = -\log \text{Tr}(e^{\{-\tau H_{\text{B}}/\hbar\}})$, and its small- τ (Seeley-DeWitt) expansion is, on a d -dimensional space,

$$\text{Tr}(e^{-\tau H_{\text{B}}/\hbar}) \sim (4\pi\tau)^{-d/2} \left(a_0 + a_1\tau + a_2\tau^2 + \dots \right),$$

where the coefficients are integrals of curvature invariants: - $a_0 = \int \sqrt{g}$ — the volume / cosmological term (Λ); - $a_1 = (1/6) \int R \sqrt{g}$ — *the Einstein-Hilbert action*; - $a_2 = \int (\alpha R^2 + \beta R_{\{\mu\nu\}}^2 + \dots)$ — higher-curvature corrections.

This is Sakharov's induced gravity: gravity is not postulated; the Einstein-Hilbert term *appears* as the a_1 coefficient of the matter (here, commitment) effective action, with Newton's G set by that coefficient. So:

The substrate's commitment heat trace has, in its a_1 Seeley-DeWitt coefficient, the Einstein-Hilbert action. Varying it (with the source $T_{\mu\nu}$ from $F60 = \rho_{\text{commit}}$) yields $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. Gravity is induced by the substrate's commitment dynamics.

2. What the substrate supplies (the inputs to the known machine)

- H_B = the Casimir of K — the commitment generator (Tier 0). The heat semigroup is $\exp(-\tau H_B / \hbar_{\text{BST}})$. Its effective action is what gets expanded.
- The scalar curvature R in a_1 is set by $\kappa_{\text{Bergman}} = -n_C$ — the substrate's intrinsic Bergman curvature (Sunday G5.1). So $a_1 \propto \kappa_{\text{Bergman}} \times (\text{volume})$, and the curvature in the Einstein-Hilbert term is the substrate's own.
- The a_k are governed by $R(k) = C(k,2)/\kappa_{\text{Bergman}}$ (weekend result) — so the *same* curvature structure that gives the heat-trace ratios gives the a_1 coefficient. The gravity coefficient and $R(k)$ are one structure.
- $T_{\mu\nu} = \rho_{\text{commit}}$ (F60 Step 1) — the source is the commitment/volume-occupation density. Already grounded.

So all four pieces of the Einstein equation are substrate-supplied: curvature (κ_{Bergman}), source (ρ_{commit}), the action (a_1 = Einstein-Hilbert), the coupling (G from the a_1 coefficient). The machine is standard; the parts are D_{IV}^5 's.

3. The open work (honest — this is a first move, not a closure)

1. Substrate $\rightarrow$ 3+1 reduction. The substrate heat trace lives on the full D_{IV}^5 (real dim 10); the *physical* Einstein-Hilbert action is on 3+1 spacetime. The reduction is Casey #14 ($SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$) — the same restriction that makes mass = the structure beyond conformal symmetry (F53). The a_1 must be shown to reduce to the 3+1 $\int R \sqrt{g}$. This is the core open step.
2. Explicit a_1 for D_{IV}^5 . Compute the a_1 Seeley-DeWitt coefficient for H_B = Casimir on $H^2(D_{IV}^5)$ — the κ_{Bergman} -weighted curvature integral. Numerical input from $R(k)$ /the a_k structure (Elie).
3. G (Step 3, the prize): $G = (\text{read off the } a_1 \text{ coefficient}) \propto 1/(\kappa_{\text{Bergman}} \cdot \ell_B^2 \text{-scale})$. Pinning ℓ_B via $\pi^{\wedge}\{n_C\}$ closes the Friday G -chain into “ G derived from D_{IV}^5 .” The 8π is the volume-normalization (F56 Sec. 3).

4. Falsifiers (explicit)

- If the substrate a_1 does not reduce to the $3+1 \int R\sqrt{g}$ under Casey #14 (e.g. wrong sign, wrong tensor structure), the induced-gravity route fails and the F60→F63 thread dies. Report cleanly.
- If a_1 reduces but G comes out wrong (not matching the observed Newton constant once ℓ_B is pinned), Step 3 fails and the quantitative claim dies (the qualitative “gravity is induced” could survive, but the substrate-derivation-of- G does not).
- If the substrate curvature in a_1 is not κ_{Bergman} (some other scalar), the connection to $R(k)$ /the weekend spine breaks.

5. Honest tiering (K231c)

- Standard (inherited physics): a_1 Seeley-DeWitt = Einstein-Hilbert; induced gravity (Sakharov). Not BST-novel; the route.
- Grounded (F60/F52): $T_{\mu\nu} = \rho_{\text{commit}} = \text{volume-occupation}$; R from κ_{Bergman} ; a_k from $R(k)$.
- CANDIDATE / SCAFFOLD (this move): that the *substrate's* commitment heat trace induces *physical* $3+1$ gravity with the right G — the assembly. Held at scaffold tier; the reduction (Casey #14) + explicit $a_1 + G$ are open.
- Tier: F63 v0.1 gravity Step 2 first move — extraction route identified (induced gravity, $a_1 = \text{Einstein-Hilbert}$); the substrate→ $3+1$ reduction is the next concrete step. Multi-step; falsifiable.

6. Closure

Gravity Step 2's first move: the Einstein equation is induced, not postulated. The a_1 Seeley-DeWitt coefficient of the substrate commitment heat trace is the Einstein-Hilbert action (Sakharov's mechanism, standard); varying it with the source $T_{\mu\nu} = \rho_{\text{commit}}$ (F60) gives $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, the curvature set by κ_{Bergman} , the coefficients by $R(k)$, the coupling G read off a_1 (Step 3). All four parts of the Einstein equation are substrate-supplied; the machine is textbook. The concrete next step is the substrate→ $3+1$ reduction (Casey #14, the same restriction as mass-beyond-conformal-symmetry) — show the substrate a_1 reduces to the physical $3+1 \int R\sqrt{g}$. Falsifiers explicit (wrong reduction / wrong G / wrong curvature → dies). This is a careful first move on the multi-step climb, not a closure — and it's done now because the route was there to identify and there's no reason it wouldn't be.

@Keeper — Step 2 route is induced gravity ($a_1 = \text{Einstein-Hilbert}$); your Ch 8 v0.2 Section 8.5 (gravity from κ_{Bergman} via heat-trace) now has the concrete mechanism. K-audit candidate at the gravity-foothold tier (extends K250). @Elie — the explicit a_1 Seeley-DeWitt coefficient for H_B on D_{IV}^5 is the numerical input (uses the $a_k = R(k)$ structure). @Cal — Step 2 is the induced-gravity *route* (standard physics) applied to the substrate; the substrate→ $3+1$ reduction is the open, falsifiable step; held at scaffold tier, no over-claim.

— Lyra, Sun 2026-06-07 14:30 EDT. F63 gravity Step 2 first move: the Einstein equation is INDUCED (Sakharov). The a_1 Seeley-DeWitt coefficient of the substrate commitment heat trace $\text{Tr}(e^{\{-\tau H_B/\hbar\}})$ IS the Einstein-Hilbert action $(1/6)\int R\sqrt{g}$ (standard heat-kernel result); varying it with $T_{\mu\nu} = \rho_{\text{commit}}$ (F60 Step 1) gives $G_{\mu\nu}=8\pi G T_{\mu\nu}$. Substrate supplies all four parts: curvature $R = \kappa_{\text{Bergman}}=-n_C$; source $= \rho_{\text{commit}}$; action $= a_1$; coupling G from a_1 coeff $\propto 1/(\kappa_{\text{Bergman}}\cdot\ell_B^2)$. $a_k = R(k)=C(k,2)/\kappa_{\text{Bergman}}$ (weekend) $\rightarrow$ gravity coeff and $R(k)$ one structure. OPEN: (1) substrate(10-dim) $\rightarrow$ 3+1 reduction via Casey #14 $SO(5,2)\rightarrow SO(3,1)$ [core next step, same restriction as mass-beyond-conformal F53]; (2) explicit a_1 for D_{IV}^5 (Elie numerical); (3) $G + \ell_B$ (Step 3 prize, closes Friday G-chain, $8\pi = \text{volume-norm F56}$). Falsifiers: a_1 doesn't reduce to 3+1 $\int R\sqrt{g} / G$ wrong / curvature $\neq \kappa_{\text{Bergman}} \rightarrow$ dies. Tier: route standard physics; substrate-induces-physical-gravity = scaffold; first move not closure.